Vectors: how AI measures that two pieces of data are alike

When an assistant searches your documents, how does it find the passage that fits your question?

Mathematics for AI and Machine Learning | Session 2 of 18 | Linear algebra

KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Write a row of data as a vector, and add, scale and compare vectors by hand



Calculate the dot product (inner product), the L2 norm, the Euclidean distance and the cosine similarity



Rank six short sentences against a query with cosine similarity, in Python and in a spreadsheet

Your two hours: vectors, then the lab, then AI search

0:00	10	Recall, then a real question from AI
0:10	35	Concept: a picture, then one example by hand
0:45	10	Break
0:55	45	Lab: predict, run, then change one thing
1:40	15	Where it appears in AI, and an exit check
1:55	5	Save the work

Ideas first. Then the lab, in Python or in a spreadsheet. Then where it shows up in AI.

Three questions from last time

1. What does the symbol Σ (sigma) mean?
2. What does the logarithm of a product equal?
3. Why do AI programs add log-probabilities?

Answer aloud first. Then we check together.

Answers to the three questions from Session 1

Question	Short answer
What does the symbol Σ (sigma) mean?	Add up everything that follows it, from the start index to the stop index.
What does the logarithm of a product equal?	The sum of the logarithms: $\ln(u \times v) = \ln u + \ln v$.
Why do AI programs add log-probabilities?	A product of many small probabilities underflows to 0. A sum of logs stays an ordinary number.

If one answer surprised you, look again at the Session 1 slides before the lab.

Words we use today

	Word	What it means
	**Vector**	A list of d numbers in a fixed order. A row of data is a vector. d is its dimension.
	**Dot product (inner product)**	Multiply matching entries of two vectors, then add. The result is one number.
	**L2 norm (Euclidean norm)**	The length of a vector: the square root of the sum of its squared entries.
	**Cosine similarity**	The cosine of the angle between two vectors: 1 for the same direction, 0 at a right angle.
	**Embedding**	A vector that represents a piece of text so that similar meanings get similar vectors.

A row of data is a vector: a list of numbers in a fixed order

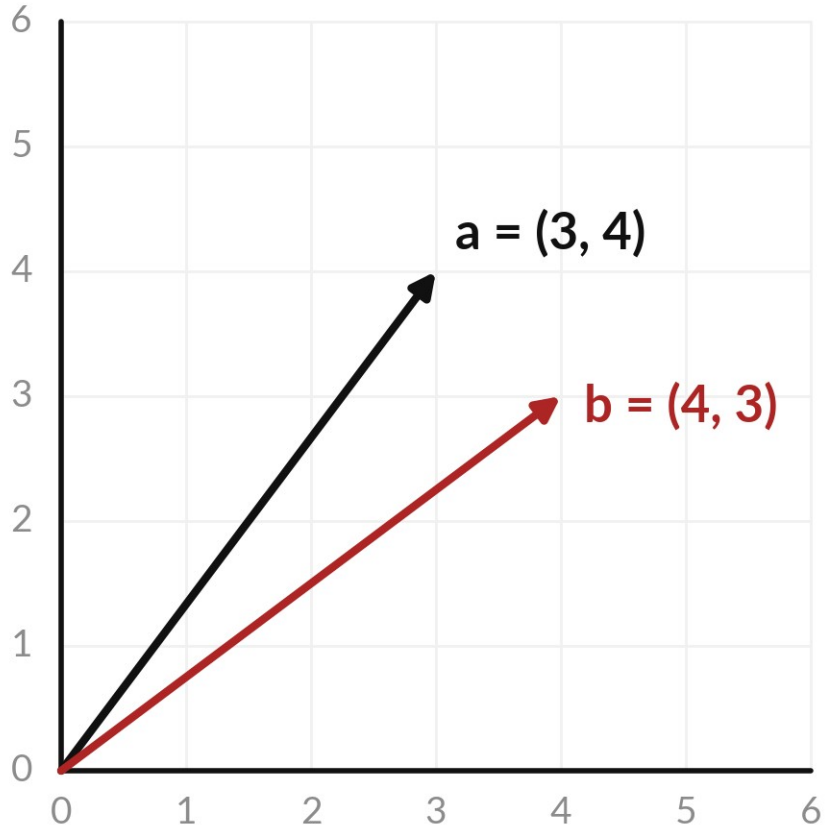
Learner	Age	Hours	Score
Amina	21	6	72
Brian	25	3	58
Chloe	19	8	85

The gold row, written as a vector:

$$\mathbf{x} = \begin{bmatrix} 21 & 6 & 72 \end{bmatrix}$$

Years, hours a week, score of 100: x = (21, 6, 72), d = 3. Written across to save space.

A vector with two numbers is an arrow from the origin to a point

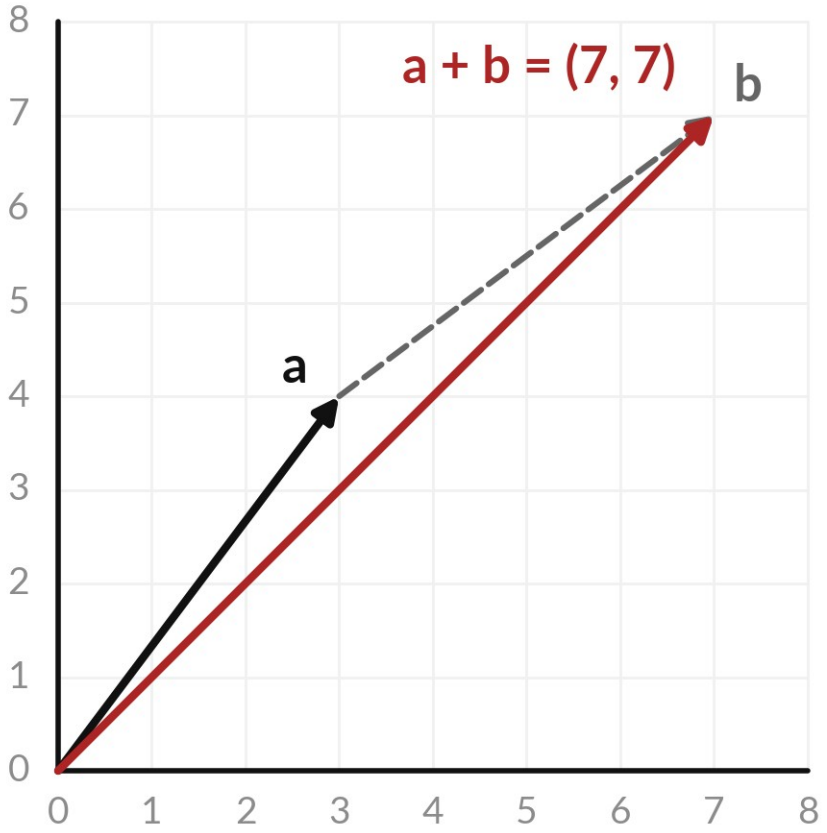


The origin is the point (0, 0). The vector a = (3, 4) is the arrow from the origin to the point 3 across and 4 up.

b = (4, 3) is a second vector: 4 across and 3 up.

With 4 or more numbers we cannot draw the arrow, but the rules stay the same.

Adding vectors places one arrow at the tip of the other

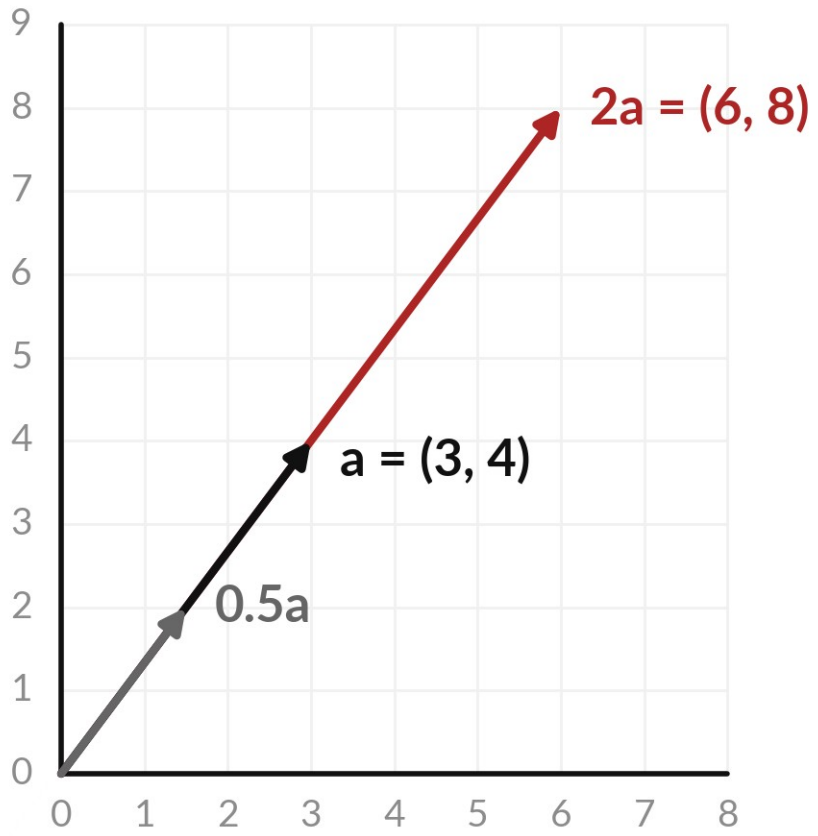


Vector addition: add the matching entries. (3, 4) + (4, 3) = (7, 7).

Draw b starting at the tip of a. The red arrow from the origin to the end is a + b.

Both vectors must have the same number of entries.

Scaling a vector by a number stretches or shrinks the arrow



Scaling: multiply every entry by one number c, called a scalar. 2 × (3, 4) = (6, 8).

A scalar above 1 stretches the arrow. A scalar between 0 and 1 shrinks it.

A positive scalar keeps the direction. A negative scalar reverses it.

Vectors add entry by entry, and a scalar multiplies every entry

Addition $$\mathbf{a} + \mathbf{b} = (a_1 + b_1,\ a_2 + b_2)$$

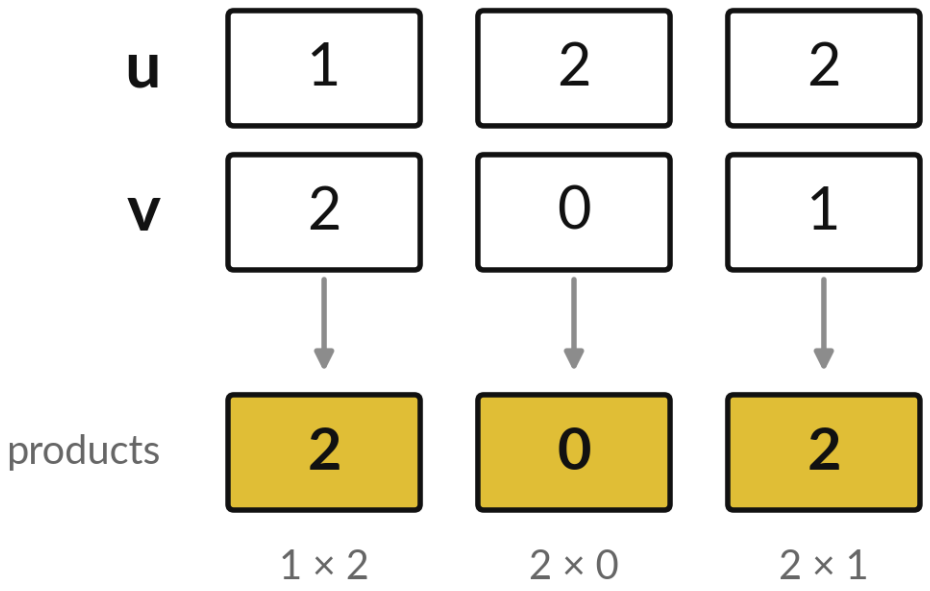
Scaling $$c\,\mathbf{a} = (c\,a_1,\ c\,a_2)$$

Our numbers $$(3, 4) + (4, 3) = (7, 7), \qquad 2\,(3, 4) = (6, 8)$$

Symbol	What it means	In our data
a and b	Two vectors with the same number of entries. Bold letters mean vectors.	a = (3, 4), b = (4, 3)
a_1, a_2	Entry 1 and entry 2 of a. The subscript is the position.	$a_1 = 3$, $a_2 = 4$
c	A single number, called a scalar.	c = 2

Written across to save space; Session 3 stands vectors up as columns. Python: a + b, 2 * a.

The dot product multiplies matching entries and adds the results



Add the products: this is the dot product of u and v

$$2 + 0 + 2 = 4$$

Dot product (also called inner product): multiply matching entries, then add.

u = (1, 2, 2) and v = (2, 0, 1).

The products are 2, 0 and 2, and they add up to 4.

The result is a single number.

Written with sigma, the dot product is a sum of d products

Dot product

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{d} u_i v_i$$

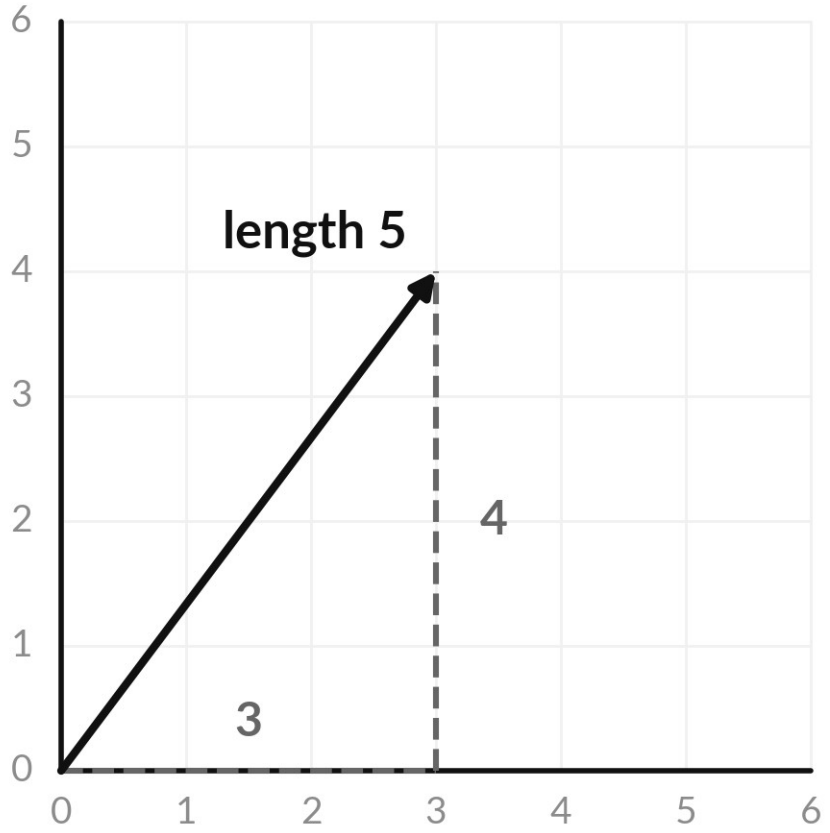
Our numbers

$$1 \times 2 + 2 \times 0 + 2 \times 1 = 4$$

Symbol	What it means	In our data
Σ and d	Add up over the d entries. d is the number of entries.	d = 3
u_i, v_i	Entry i of u and of v.	u = (1, 2, 2), v = (2, 0, 1)
u · v	The dot product: one number. The raised dot is kept for this; × multiplies two numbers.	4

Python: u @ v or np.dot(u, v). Spreadsheet: =SUMPRODUCT(u, v). The cosine slides later link this number to the angle.

The L2 norm is the length of the arrow, from the Pythagoras rule



L2 norm (Euclidean norm): the length of a vector. It is written ‖a‖.

For a = (3, 4): $3^2 + 4^2 = 9 + 16 = 25$.

The square root of 25 is 5, so ‖a‖ = 5.

A longer arrow has a bigger norm.

The L2 norm is the square root of a vector dotted with itself

L2 norm (length)

$$\|\mathbf{a}\|_2 = \sqrt{\sum_{i=1}^{d} a_i^2} = \sqrt{\mathbf{a} \cdot \mathbf{a}}$$

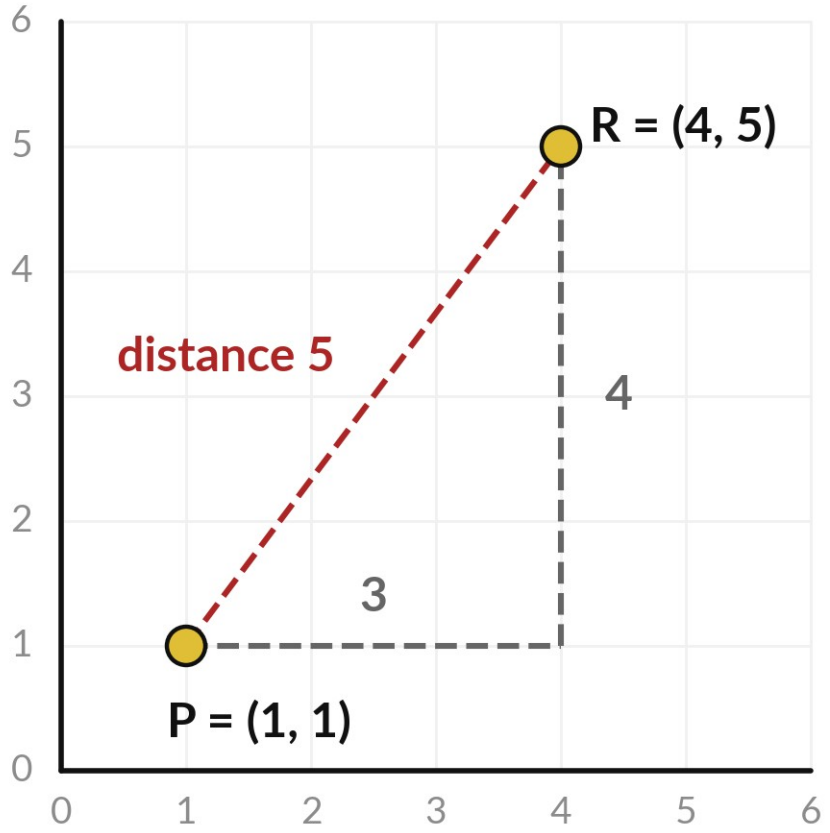
Our numbers

$$\sqrt{3^2 + 4^2} = 5, \qquad \sqrt{1^2 + 2^2 + 2^2} = 3$$

Symbol	What it means	In our data
$\|a\|_2$	The L2 norm. The subscript 2 means squares and a square root.	5 for (3, 4)
Σ and d	Add up over the d entries of a.	d = 2
a_i	Entry i of the vector a.	3 and 4
a · a	The dot product of a with itself: the squared length.	25

Python: np.linalg.norm(a). Spreadsheet: =SQRT(SUMPRODUCT(a, a)).

Euclidean distance is the length of the arrow between two points



Euclidean distance between the points P and R: the L2 norm of their difference, R − P.

R − P = (3, 4), so the distance is 5.

A small distance means the two rows are close, so they are alike.

Distance subtracts entry by entry, squares, adds and takes the root

Euclidean distance

$$\text{dist}(\mathbf{P}, \mathbf{R}) = \|\mathbf{R} - \mathbf{P}\|_2 = \sqrt{\sum_{i=1}^{d} (R_i - P_i)^2}$$

Our numbers

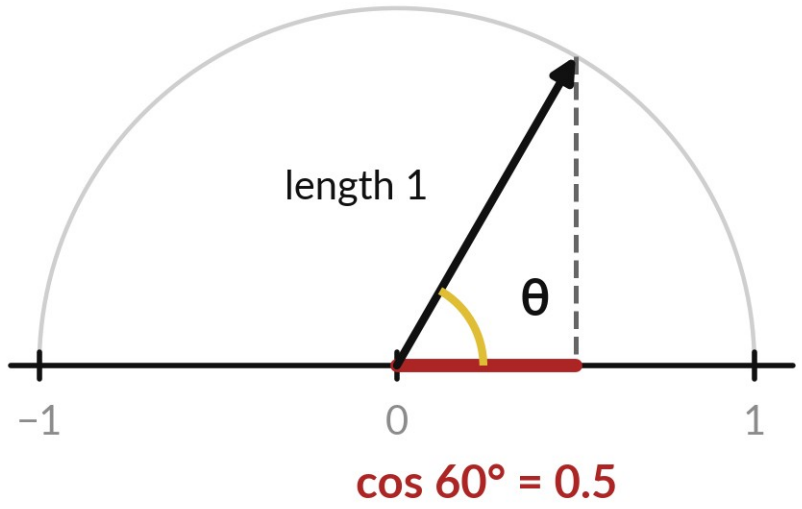
$$\sqrt{(4-1)^2 + (5-1)^2} = \sqrt{9 + 16} = 5$$

Symbol	What it means	In our data
P and R	Two points, each written as a vector with d entries.	P = (1, 1), R = (4, 5)
P_i, R_i	Entry i of P and of R.	$R_1 - P_1 = 3$, $R_2 - P_2 = 4$
Σ and d	Add up over the d entries.	d = 2
dist(P, R)	The distance: the length of the arrow from P to R.	5

Python: np.linalg.norm(R - P). Spreadsheet: =SQRT(SUMXMY2(P, R)). The letter q is kept for the query vector.

The cosine of an angle tells how far an arrow of length 1 reaches along the horizontal

An arrow of length 1 at angle θ = 60°



angle	radians	cosine
0°	0	1
60°	1.047	0.5
90°	1.571	0
180°	3.142 (π)	−1

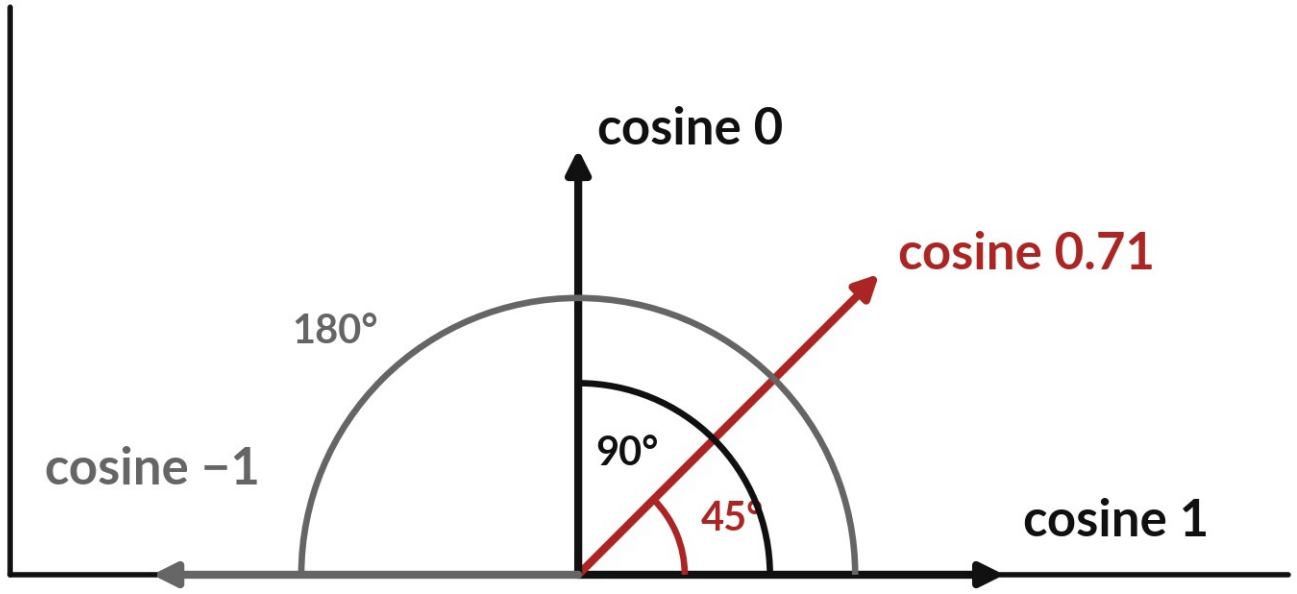
Trigonometry refresher. θ (theta) is an angle. An arrow of length 1 at angle θ reaches cos θ sideways.

cos θ: 1 at 0°, 0 at 90°, −1 at 180°.

arccos goes back to the angle: ACOS, np.arccos.

Computers give radians: 180° = π ≈ 3.142. DEGREES() or np.degrees() converts.

Cosine similarity compares directions and ignores the lengths of the vectors



Cosine similarity: the cosine of the angle θ between two vectors.

Angles are measured from the horizontal arrow (cosine 1).

Same direction: 1. At right angles: 0. Opposite directions: −1.

The lengths do not matter.

Divide the dot product by both lengths and the cosine is left

Cosine similarity

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \quad \text{so} \quad \mathbf{a}\cdot\mathbf{b} = \|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta$$

a, b and u, v

$$\frac{24}{5\times 5} = 0.96 \rightarrow \theta = 16.26^\circ; \quad \frac{4}{3\times 2.236} = 0.596 \rightarrow \theta = 53.4^\circ$$

Symbol	What it means	In our data
a · b	The dot product: 3 × 4 + 4 × 3 = 24.	24
‖a‖, ‖b‖	The L2 norms of the two vectors.	5 and 5
θ	The angle between the arrows. arccos undoes cosine.	16.26 degrees

Angle: np.degrees(np.arccos(cos)) or =DEGREES(ACOS(cos)). ACOS alone gives radians.

Distance and cosine similarity answer different questions

Euclidean distance

Compares positions.

a = (3, 4) and 2a = (6, 8): distance 5.

Small means similar.

Cosine similarity

Compares directions.

The same two vectors: cosine 1.

Large means similar.

A vector stretched for any reason keeps its cosine but moves away in distance.

Common mistake: trusting the biggest dot product as the best match

Wrong

The largest dot product is the closest match.

A vector 10 times longer scores 10 times higher.

Right

Divide by both lengths first: that is cosine similarity.

Lab: picnic vector × 20. Dot product 1.6 beats 0.785; cosine stays 0.097.

Normalising scales every vector to length 1 (a unit vector). Then dot product equals cosine.

An embedding turns a piece of text into a vector

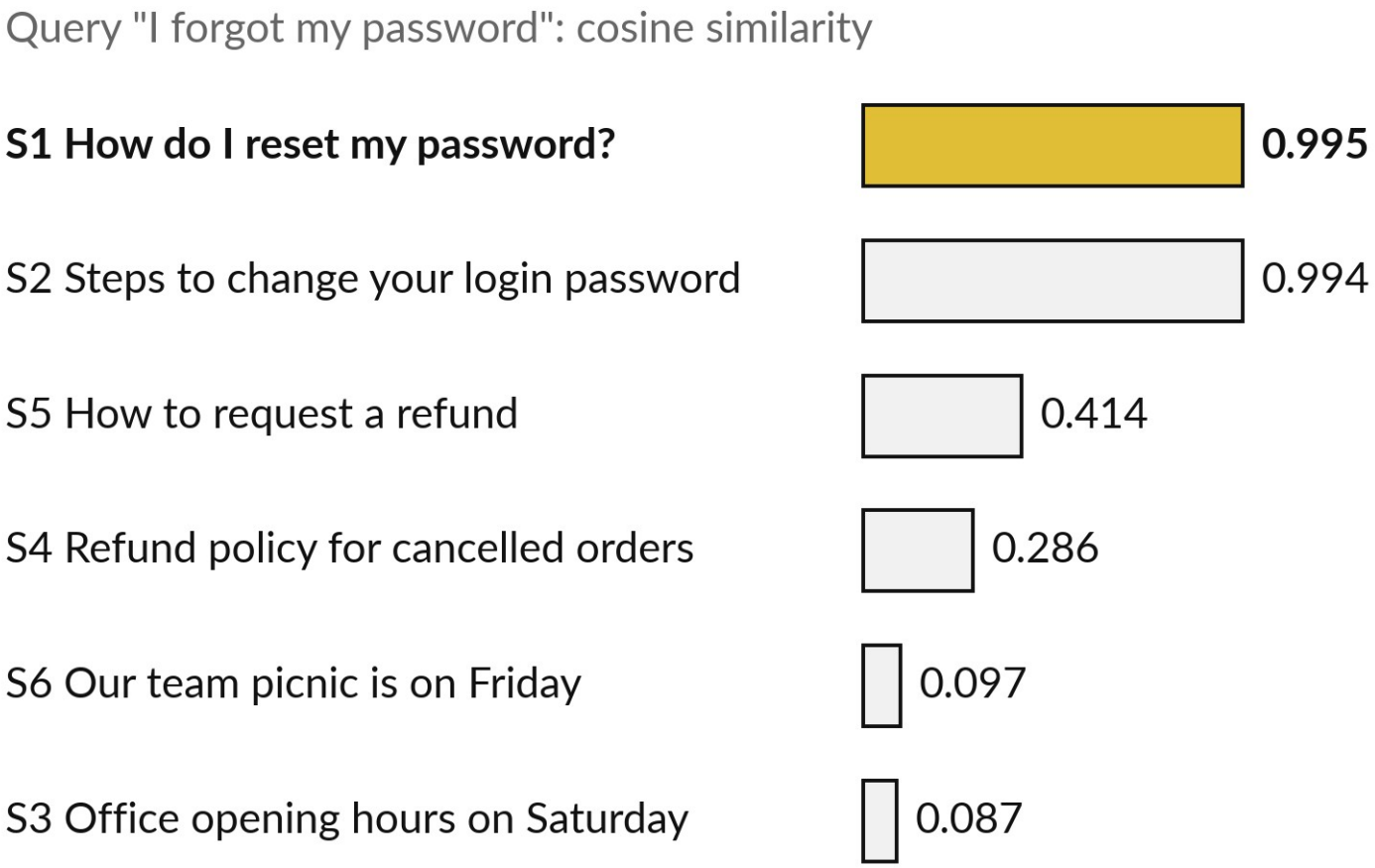
	A	B	C	D	E
	Sentence	**d1**	**d2**	**d3**	**d4**
1	S1 How do I reset my password?	0.9	0.1	0	0.1
2	S2 Steps to change your login password	0.8	0.2	0.1	0
3	S3 Office opening hours on Saturday	0	0.1	0.9	0.2
4	S4 Refund policy for cancelled orders	0.1	0.9	0.2	0
5	S5 How to request a refund	0.2	0.8	0.1	0.1
6	S6 Our team picnic is on Friday	0	0.1	0.5	0.8

An embedding is a vector that represents text, so that similar meanings get similar vectors.

Each row is one short sentence with d = 4 numbers.

These numbers are illustrative (made-up numbers). Real embeddings have hundreds of numbers.

Ranking by cosine similarity finds the sentence closest to the query



The query is also turned into a vector: (0.85, 0.15, 0.05, 0.05). Compare it with every sentence.

S1 and S2, the password sentences, score 0.995 and 0.994.

The picnic and opening hours score below 0.1.

Read the recipe before you run it: dot product, norm and cosine

```
dot = np.sum(u * v)
norm_a = np.sqrt(np.sum(a * a))
cos = a @ b / (norm_a * norm_b)
norms = np.linalg.norm(E, axis=1)
norm_q = np.linalg.norm(q)
sims = E @ q / (norms * norm_q)
best = np.argmax(sims)
```

In plain words

Line 1: dot product. Spreadsheet: SUMPRODUCT.

Line 2: L2 norm. Spreadsheet: SQRT(SUMPRODUCT(a, a)).

Line 3: cosine of a and b.

Lines 4 to 6: one norm per row of E (axis=1), the norm of q, all six cosines.

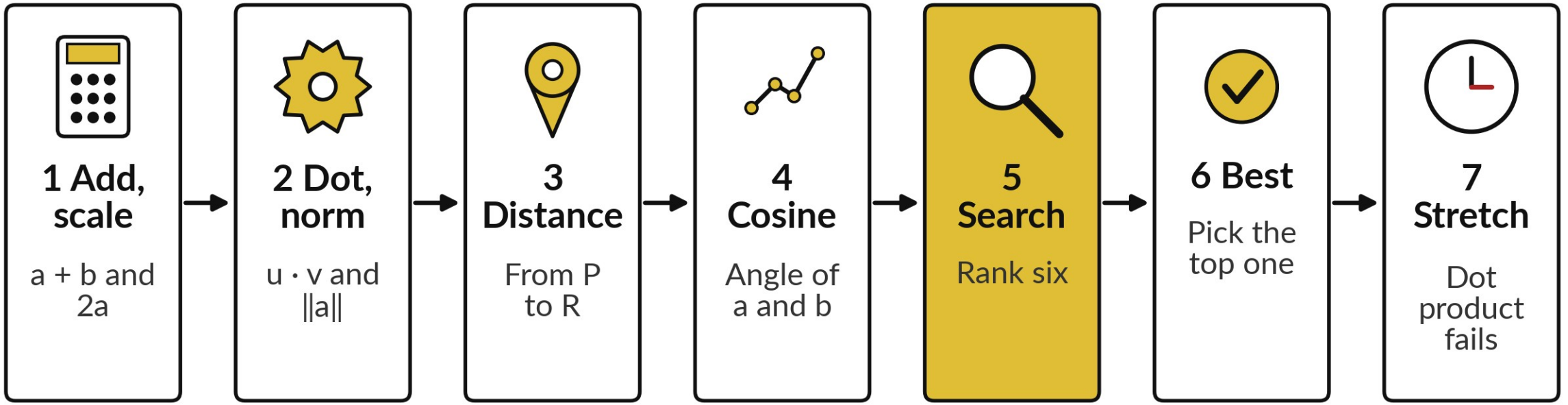
Line 7: position of the largest. Spreadsheet: =INDEX(A4:A9, MATCH(MAX(H4:H9), H4:H9, 0)).



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has seven steps, from a hand calculation to a search



Predict, run, read, change. Step 7 stretches one vector by 20 to see the dot product fail.

Check your numbers before you move on

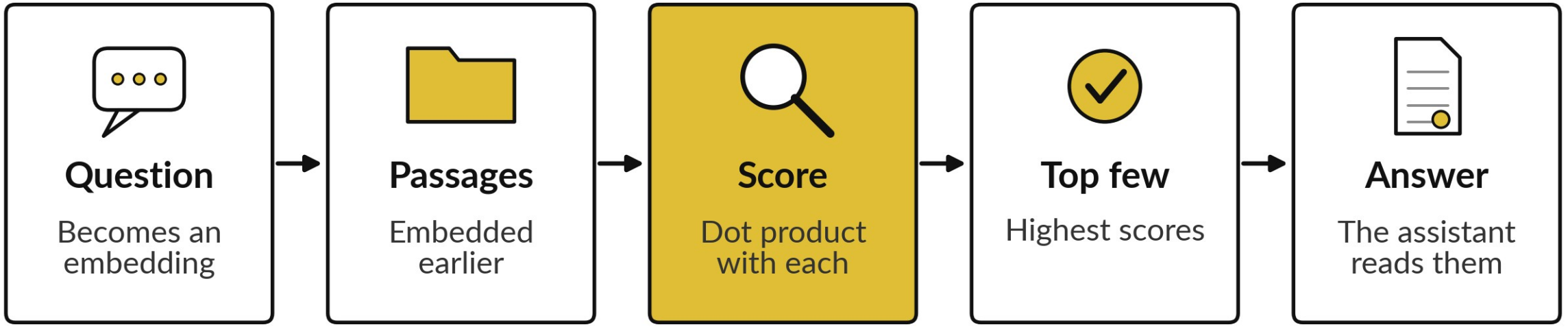
	What you compute	You should get
	a + b for a = (3, 4), b = (4, 3)	(7, 7)
	Dot product of u = (1, 2, 2) and v = (2, 0, 1)	4
	L2 norm of a = (3, 4)	5
	Distance from P = (1, 1) to R = (4, 5)	5
	Cosine similarity of a and b	0.96
	Angle between a and b, in degrees	16.26
	Cosine of the query with sentence 1	0.995
	Best match for the query	S1 How do I reset my password?

If your number differs, find the step that went wrong. Do not change a number to match.

Four messages you may meet, what they mean and what to do

	What you see	What it means	What to do
	**ValueError: operands could not be broadcast together**	You combined two vectors with different numbers of entries.	Print a.shape and b.shape. They must match.
	**My cosine is larger than 1**	A division is missing, or the wrong norm was used.	Divide by both norms, each from its own vector.
	**#VALUE! in a spreadsheet**	The two ranges in SUMPRODUCT have different sizes.	Select ranges with the same number of cells.
	**The angle shows 0.284**	ACOS and np.arccos give radians.	Wrap it: DEGREES(ACOS(...)) or np.degrees(...).

Where this appears in AI: finding the right passage



RAG (Lewis et al., 2020) scores by dot product; for length-1 vectors that is the cosine.

Where this appears in AI: check an assistant with your numbers

8 minutes Hands-on activity

Ask any AI assistant: "What is the cosine similarity of (3, 4) and (4, 3)?"

Use only public maths. Never paste confidential data.

Compare its answer with your lab number, 0.96.

Ask it to show the dot product and lengths. Do they match 24, 5 and 5?

Ask for the angle: about 16.26 degrees?

You keep

A note: the assistant answer, your lab number, and whether they agree.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**What is the dot product of (2, 3) and (4, 1)?**	11, because $2 \times 4 + 3 \times 1 = 8 + 3$.
	**What is the L2 norm of (6, 8)?**	10, because 36 + 64 = 100 and the square root of 100 is 10.
	**When is ranking by the raw dot product risky, and what fixes it?**	When vector lengths differ: a long vector can win on length alone. Divide by both lengths (cosine), or normalise every vector to length 1 first.

Save your lab file and add four formulas to your sheet

What you keep from today

Keep your saved notebook or workbook, with every check on OK. Keep the one idea of the day: cosine similarity compares the direction of two vectors, and that is how AI finds a passage that means the same.

Our question today

When an assistant searches your documents, how does it find the passage that fits your question?

Next session

Session 3: Matrices. A table of numbers that can also move and reshape vectors.

Take away

Your lab file
Saved, all checks OK

The idea
cosine = a·b / (‖a‖ ‖b‖)

Formula sheet
dot, norm, distance, cosine

Next session
Matrices